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(54) **VEHICLE UPFITTER POWER CONNECTION MANAGEMENT STRATEGIES**

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B60R 16/03 (2006.01)

(52) **U.S. Cl.**
CPC ***B60R 16/03*** (2013.01); ***G01R 31/08*** (2013.01)

(58) **Field of Classification Search**
CPC B60R 31/08; G01R 31/08
See application file for complete search history.

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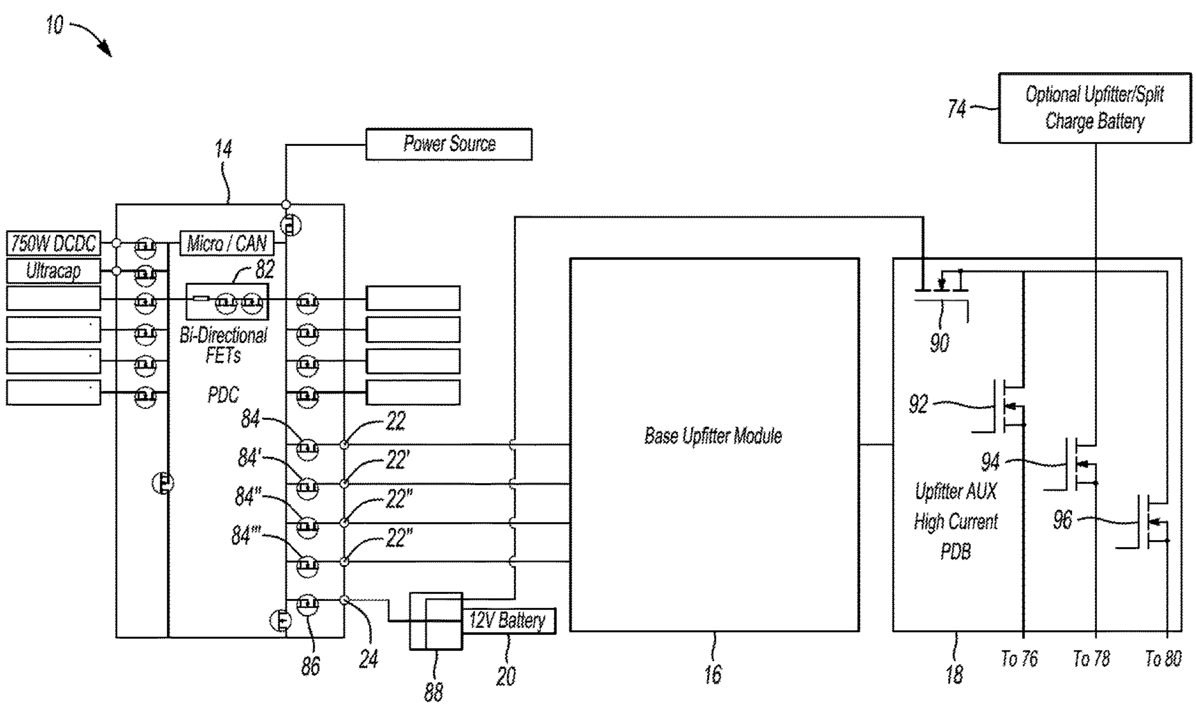
(57) **ABSTRACT**

One or more controllers may, after activation of a vehicle, sequentially pulse field effect transistors of a power distribution module of the vehicle that are connected to inputs of an upfitter power module, and preclude operation of the upfitter power module responsive to, for at least two of the field effect transistors, electrical parameters resulting from the corresponding pulses being reported for a same one of the inputs.

20 Claims, 3 Drawing Sheets



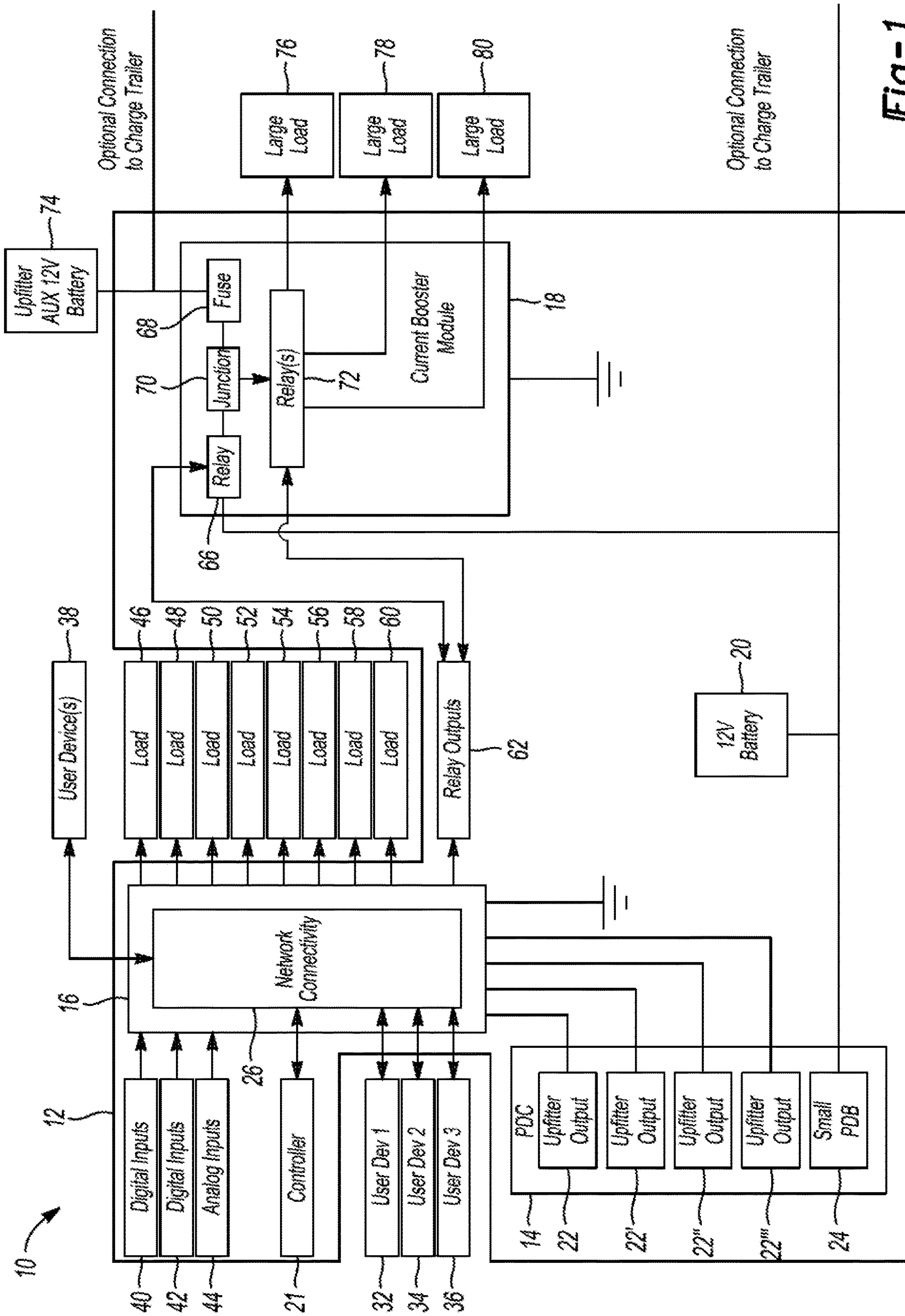
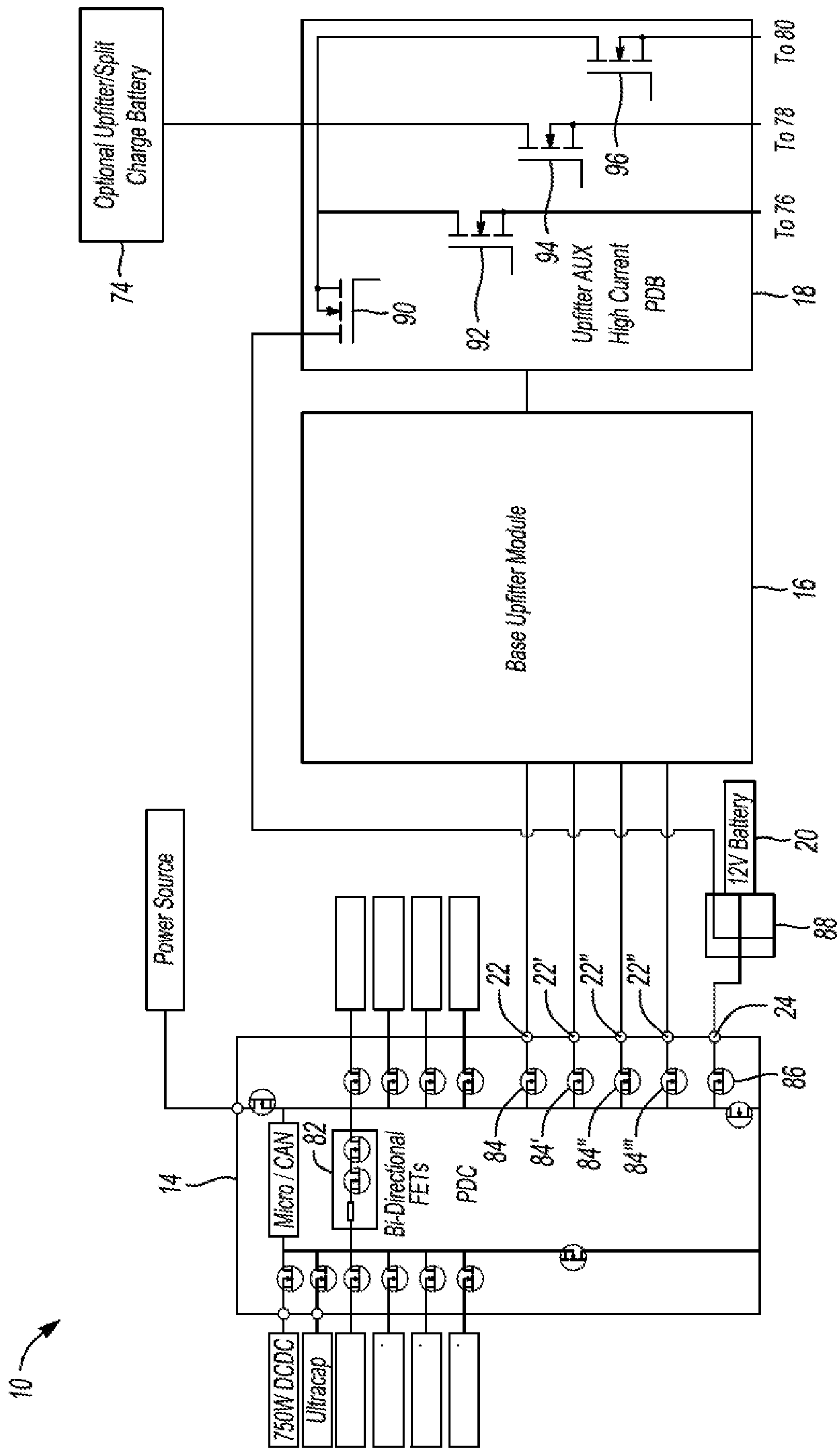

10
Optional Connection to Charge Trailer
Optional Connection to Charge Trailer
Upfitter AUX 12V Battery
74
Fuse
68
Junction
70
Relay
66
Relay(s)
72
Current Booster Module
18
Large Load
76
Large Load
78
Large Load
80
User Device(s)
38
Load
46
48
50
52
54
56
58
60
Relay Outputs
62
12V Battery
20
Network Connectivity
16
26
12
Digital Inputs
Digital Inputs
Analog Inputs
40
42
44
Controller
21
User Dev 1
User Dev 2
User Dev 3
32
34
36
PDC
14
Upfitter Output
22
22'
22''
22'''
Small PDB
24

Fig-1



Optional Upfitter/Split Charge Battery
74
To 80
To 78
To 76
96
94
92
90
Upfitter AUX High Current PDB
18
Base Upfitter Module
16
Power Source
22
22'
22"
22"
24
12V Battery
20
88
86
14
Micro / CAN
82
Bi-Directional FETs
PDC
84
84'
84"
84'"
750W DCDC
Ultracap
10

Fig-2

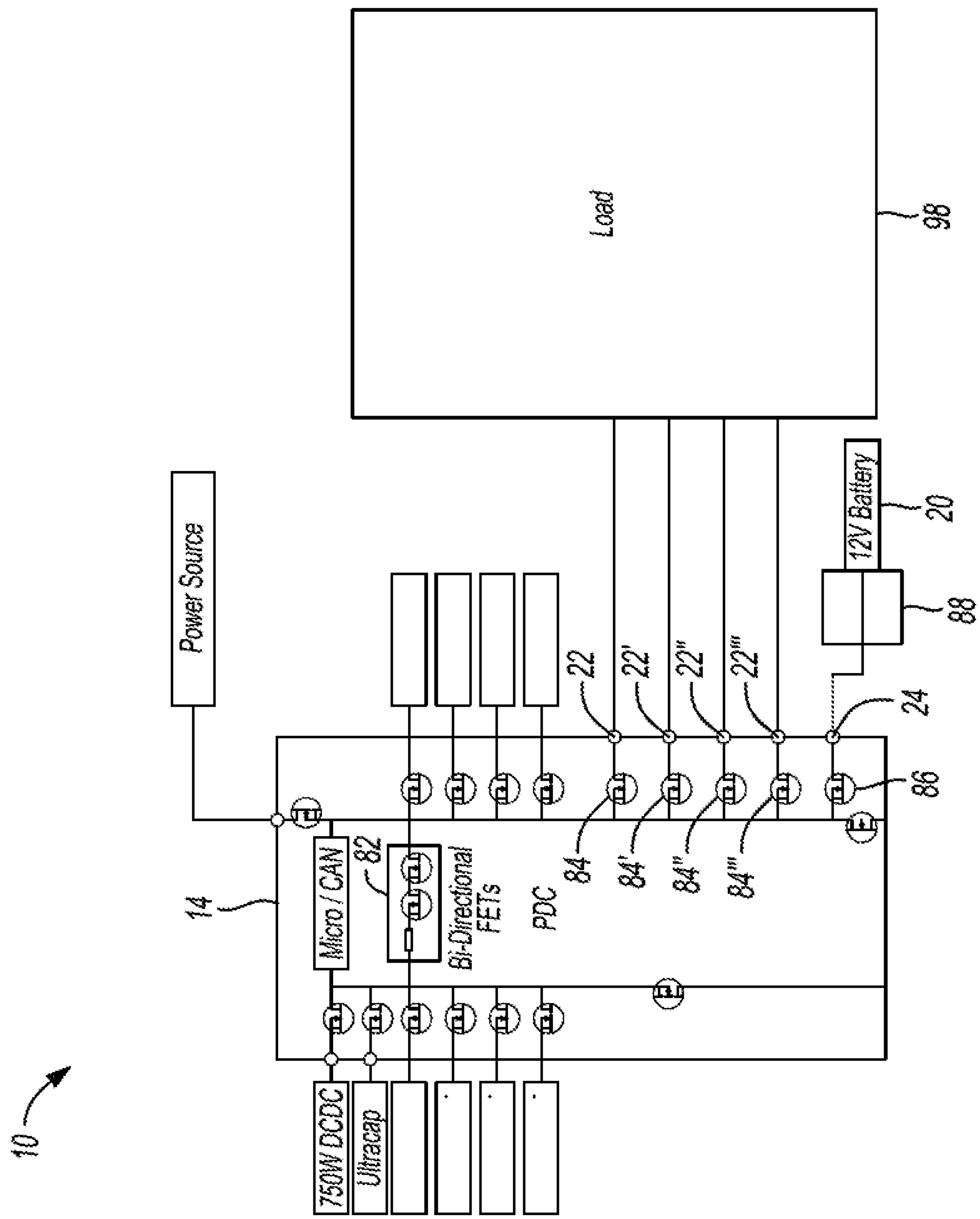

Load
98
Power Source
12V Battery
20
88
22
22'
22''
22'''
24
86
84
84'
84''
84'''
14
Micro / CAN
82
Bi-Directional FETs
PDC
750W DCDC
Ultracap
10

Fig-3

VEHICLE UPFITTER POWER CONNECTION MANAGEMENT STRATEGIES

TECHNICAL FIELD

This disclosure relates to automotive power systems.

BACKGROUND

An automotive vehicle may include an energy storage device, such as a battery. This energy may be made available to electrical and electronic components of the vehicle and outside the vehicle. After-market plowing equipment, for example, may be powered by energy from a battery. Such connections with the battery may be facilitated by a so-called upfitter module.

SUMMARY

A vehicle has a power distribution module including a plurality of field effect transistors and a plurality of corresponding outputs that are each able to be connected with one of a plurality of inputs of an upfitter power module, and one or more controllers that, after activation of the vehicle, sequentially pulse the field effect transistors, and preclude operation of the upfitter power module responsive to, for one of the field effect transistors, electrical parameters resulting from the corresponding pulse being reported for more than one of the inputs.

A method includes, while a plurality of field effect transistors of a power distribution module of a vehicle are passing currents to an upfitter load, sequentially toggling each of the field effect transistors, and responsive to other of the field effect transistors experiencing an increase in current flow during the toggling, deactivating the field effect transistors.

A power control system of a vehicle has one or more controllers that, after activation of the vehicle, sequentially pulse field effect transistors of a power distribution module of the vehicle that are connected to inputs of an upfitter power module, and preclude operation of the upfitter power module responsive to, for at least two of the field effect transistors, electrical parameters resulting from the corresponding pulses being reported for a same one of the inputs.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. **1** through **3** are block diagrams of an automotive power system and upfitter modules/loads.

DETAILED DESCRIPTION

Embodiments are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments may take various and alternative forms. The figures are not necessarily to scale. Some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art.

Various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations

and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

Automotive vehicles typically include a power supply to facilitate functioning of various electrical and electronic components. Several power supply architectures are commonly used in automotive applications, including centralized, decentralized, and hybrid power supply architectures.

Centralized power supply architectures can include a single power supply unit that provides power to all electrical and electronic components in the vehicle. This architecture may be simple and easy to install. In some circumstances however, it may not be suitable for vehicles with high power requirements or those that have different voltage levels for different components.

Decentralized power supply architectures can include multiple power supply units located near the components they power. This architecture may be more flexible than centralized power supply architectures as it permits different voltage levels to be provided to different components. It can also be efficient by reducing the power loss that occurs when power is transmitted over relatively long distances. Decentralized power supply architectures, however, can be more complex with the potential increase in number of connections between power supply units and corresponding components.

Hybrid power supply architectures can be a combination of centralized and decentralized power supply architectures: A central power supply unit provides power to some electrical and electronic components, while other components are powered by decentralized power supply units. This architecture, in certain applications, may provide a balance between flexibility and efficiency.

The use of upfitter power modules can affect the power supply systems of automotive vehicles. Upfitter modules, in some examples, are modular components that can be added to a vehicle to provide additional functionality, such as lighting, power distribution, or electrical power management. An upfitter module that provides lighting and power distribution, for example, can be installed in a vehicle, allowing it to be used as a work vehicle, such as a service truck or construction vehicle. Additionally, upfitter modules can be designed to meet specific requirements, such as providing power to a specific component or group of components.

Several factors can be considered when selecting an upfitter module, including the power requirements of the component or group of components that the upfitter module will power, the voltage level required, and the environmental conditions in which the upfitter module will be used.

Some upfitter modules may not support medium or high current applications, may not be expandable, may not provide feedback to the vehicle to manage effects of high current loads, and/or do not support added video cameras, etc.

Certain upfitter modules have taken the form of switches mounted in an over-head compartment and relays in an add-on engine compartment power distribution box. In some arrangements, wires are fed to the engine compartment power distribution box, but not connected to relays. Screens may be used to support touch drive of field effect transistors that control loads.

Referring to FIG. **1**, an example automotive power system **10** for a vehicle **12** includes a power distribution center **14**, an upfitter module **16**, an upfitter current booster module **18**, a 12V battery **20**, and a controller **21**.

The power distribution center **14** includes a plurality of upfitter outputs **22**, **22'**, **22"**, **22'''** (e.g., two 20 A field effect transistor controlled outputs and two 30 A field effect transistor controlled outputs, four 25 A field effect transistor controlled outputs, etc.) and an interconnector power distribution box **24** (e.g., a 200 A field effect transistor controlled output). The respective ratings of these field effect transistors thus define what their limits are for current flow therethrough. The upfitter outputs **22**, **22'**, **22"**, **22'''** are connected with the upfitter module **16** at corresponding inputs. The interconnector power distribution box **24** is connected with the 12V battery **20**. Other arrangements, with differing numbers of outputs, etc., are of course also contemplated.

The upfitter module **16** is grounded and includes network connectivity **26**, which is connected with a plurality of user devices **32**, **34**, **36**, **38**. The upfitter module **16** is configured to receive digital inputs **40**, **42** and an analog input **44**, and to provide current to loads **46**, **48**, **50**, **52**, **54**, **56**, **58**, **60** in this example. The upfitter module **16** is further configured to provide current to relay outputs **62**. Multiple such upfitter modules can be so connected (e.g., daisy chained) to obtain as many inputs/outputs as necessary as mentioned above.

The upfitter current booster module **18** is grounded and includes a relay **66**, a fuse **68**, a junction **70**, and a plurality of relays **72**. The junction **70** is connected between the relay **66**, fuse **68**, and relays **72**. The relay **66** is connected with the interconnector power distribution box **24**. The fuse **68** is connected with an auxiliary 12V battery **74**. The relays **72** are configured to provide current to a plurality of large loads **76**, **78**, **80**.

In the above described high current architecture, the upfitter current booster module **18**, which can be stamped track relays controlled by the upfitter module **16**, is powered from a power distribution center output different from the upfitter module **16**. All the expansion modules receive their power from a 100 A output from the power distribution center **14**. As current usage approaches the 100 A limit, the operator can be notified (e.g., by screen or phone-as-a-key) via, for example, alerts generated and forwarded by the controller **21**. The upfitter module **16** tracks and reports on loads to the controller **21**, which can facilitate notification of the operator.

Optional connections to a charge trailer are shown for the 200 A output. Communication with the charge trailer can be, for example, controller area network or Ethernet. This allows the upfitter to exceed the 200 A continuous power distribution center limits.

Referring to FIG. **2**, the power distribution center **14** further includes, in this example, a plurality of bidirectional field effect transistors **82**, 100 A (in the aggregate) capable field effect transistors **84**, **84'**, **84"**, **84'''** (continuous), and a 200 A capable field effect transistor **86** (continuous). The field effect transistors **84**, **84'**, **84"**, **84'''** are respectively connected between the bidirectional field effect transistors **82** and upfitter outputs **22**, **22'**, **22"**, **22'''**. The 200 A capable field effect transistor **86** is connected between the bidirectional field effect transistors **82** and the interconnector power distribution box **24**. The power distribution center **14** also includes other field effect transistors connected between the bidirectional field effect transistors **82** and other vehicle components (e.g., a DC/DC converter, an ultra-capacitor, other power sources, etc.).

The automotive power system **10** may further include a power distribution box **88**. The power distribution box **88** may be packaged on top of or next to the 12V battery **20**. It may serve as an interconnection point for the upfitter current booster module **18**, 12V battery **20**, interconnector power distribution box **24**, and any jump start post.

The upfitter current booster module **18** further includes a plurality of field effect transistors **90**, **92**, **94**, **96**. The field effect transistor **90** is connected between the power distribution box **88** and field effect transistors **92**, **94**, **96**. The field effect transistor **92** is connected between the large load **76** and field effect transistor **90**. The field effect transistor **94** is connected between the auxiliary 12V battery **74**, large load **78**, and field effect transistor **90**. The field effect transistor **96** is connected between the large load **80** and field effect transistor **90**. The field effect transistor **90** is thus arranged to isolate the large load **76**, **78**, **90** from negative transient loads. These loads may be some of the loads that are assigned priority in the event shedding becomes necessary as described above.

To achieve the 100 A continuous power, in the aggregate, from the upfitter outputs **22**, **22'**, **22"**, **22'''**, they should not be connected together in parallel as they are configured to work separately. That is, if the field effect transistors **84**, **84'**, **84"**, **84'''** are connected in parallel, the full 100 A in the aggregate output may not be achieved, and there may be unbalanced current throughput for each.

Because the upfitter module **16** is configured to measure corresponding voltages via, for example, common voltage sensors at the inputs, upon the vehicle **12** being activated and/or before operation of the upfitter module **16** is permitted (enablement), each of the field effect transistors **84**, **84'**, **84"**, **84'''** may be individually turned on/off at the command of the controller **21** while voltages at the inputs to the upfitter module **16** are checked to determine if any lines have been tied together. As just mentioned, these checks can include pulsing (turning on/off or injecting a small current pulse) each of the field effect transistors **84**, **84'**, **84"**, **84'''**. The controller **21** can then detect, for each of the field effect transistors **84**, **84'**, **84"**, **84'''**, whether voltages at multiple of the inputs are detected (one output to many inputs) based on data reported by the upfitter module **16**. The controller **21** can also detect, for two or more of the field effect transistors **84**, **84'**, **84"**, **84'''**, whether voltages at a same one of the inputs is detected (many outputs to one input). If so, the controller **21** may preclude operation of the upfitter module **16** (e.g., prevent the upfitter module **16** from facilitating power flow) and/or alert a user via, for example, an infotainment screen, to reconnect so as to not have parallel connections. If not (i.e., voltage is detected at only one of the inputs for a pulsing of a given one of the field effect transistors **84**, **84'**, **84"**, **84'''**), the controller **21** may confirm the power distribution center connections are appropriately connected (not connected in parallel) to the upfitter module **16**, and permit operation of the same.

Referring to FIG. **3**, there may be circumstances in which the field effect transistors **84**, **84'**, **84"**, **84'''** are directly powering a load **98**. In such circumstances, electrical parameter feedback (e.g., voltage measurements) from the load **98** may be unavailable. If the field effect transistors **84**, **84'**, **84"**, **84'''** are connected in parallel and powering the load **98** however, turning off one of the field effect transistors **84**, **84'**, **84"**, **84'''** would result in a current rise in other of the field effect transistors **84**, **84'**, **84"**, **84'''**. Typical current sensors associated with the field effect transistors **84**, **84'**, **84"**, **84'''** can be used to measure corresponding currents therethrough after each of the field effect transistors **84**, **84'**, **84"**, **84'''** has been toggled off at the command of the controller **21**. This current information may be reported to the controller **21**. If the same detects the current rise responsive to the toggling, the controller **21** may disable (deactivate) particular ones of field effect transistors **84**, **84'**, **84"**, **84'''** determined to be connected in parallel to preclude power flow therethrough. The controller **21** may further issue alerts/instructions as mentioned above to the user to reconnect the load **98** and not have parallel connections, etc.

Alternatively, if the controller **21** detects the current rise responsive to the toggling, it may further examine whether, for those of the field effect transistors **84**, **84'**, **84"**, **84'''** determined to be connected in parallel, a difference in current magnitudes exceeds some calibrated threshold, which can be determined via testing, simulation etc. If yes, the controller **21** may disable the subject field effect transistors **84**, **84'**, **84"**, **84'''**. If no, the controller **21** may maintain operation of the field effect transistors.

If the current magnitudes for each path are relatively close in magnitude (e.g., less than the calibrated threshold), this will indicate that the field effect transistor resistances/impedances are also relatively close in value and the power distribution center/upfitter module capability will be minimally affected. If this check indicates a difference in current/impedance (e.g., greater than the calibrated threshold), this will indicate a derated power distribution center/upfitter module capability. Based on a determination that the connections are being run in parallel, the controller **21** may also communicate to the user this arrangement could result in reduced capability and inform the user of the expected performance reduction based on the measurements. The controller **21** can also instruct the user to rewire the upfitter devices based on recommendations and/or connect the installer to a dealership technician.

If any loads are found to be tied together as mentioned above, an alert may be displayed regarding non-optimal connection to power, and/or based on the current flowing through each of the field effect transistors **84**, **84'**, **84"**, **84'''**, an estimate be generated of the full power that will be reached given the connection configuration.

If the field effect transistors **84**, **84'**, **84"**, **84'''** do reach the current limit in a parallel connected situation (typically, one will reach the current limit first then be shut down immediately followed by a second to reach the current limit then be shut down, and so on), a diagnostic code can be stored and the maximum current before shutting down at the time the first of the field effect transistors **84**, **84'**, **84"**, **84'''** tripped can be shown for all field effect transistors connected in parallel.

The algorithms, methods, or processes disclosed herein can be deliverable to or implemented by a computer, controller, or processing device, which can include any dedicated electronic control unit or programmable electronic control unit. Similarly, the algorithms, methods, or processes can be stored as data and instructions executable by a computer or controller in many forms including, but not limited to, information permanently stored on non-writable storage media such as read only memory devices and information alterably stored on writeable storage media such as compact discs, random access memory devices, or other magnetic and optical media. The algorithms, methods, or processes can also be implemented in software executable objects. Alternatively, the algorithms, methods, or processes can be embodied in whole or in part using suitable hardware components, such as application specific integrated circuits, field-programmable gate arrays, state machines, or other hardware components or devices, or a combination of firmware, hardware, and software components.

While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of these disclosed materials. The terms "controller" and "controllers," for example, can be used interchangeably herein as the functionality of a controller can be distributed across several controllers/modules, which may all communicate via standard techniques.

As previously described, the features of various embodiments may be combined to form further embodiments of the invention that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics may be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes may include, but are not limited to strength, durability, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and may be desirable for particular applications.

What is claimed is:

1. A vehicle comprising:

a power distribution module including a plurality of field effect transistors and a plurality of corresponding outputs that are each configured to be connected with one of a plurality of inputs of an upfitter power module; and

one or more controllers programmed to, after activation of the vehicle, sequentially pulse the field effect transistors, and preclude operation of the upfitter power module responsive to, for one of the field effect transistors, electrical parameters resulting from the corresponding pulse being reported for more than one of the inputs.

2. The vehicle of claim **1**, wherein the one or more controllers are further programmed to, after activation of the vehicle, preclude operation of the upfitter power module responsive to, for two or more of the field effect transistors, electrical parameters resulting from the corresponding pulses being reported for a same one of the inputs.

3. The vehicle of claim **1**, wherein the one or more controllers are further programmed to enable of the upfitter power module responsive to, for each of the field effect transistors, an electrical parameter resulting from the pulse being reported for only one of the inputs.

4. The vehicle of claim **1**, wherein the one or more controllers are further programmed to, while the field effect transistors are passing currents, sequentially toggle each of the field effect transistors, and responsive to other of the field effect transistors experiencing an increase in current flow during the toggle, deactivate the field effect transistors.

5. The vehicle of claim **1**, wherein the one or more controllers are further programmed to, while the field effect transistors are passing currents, sequentially toggle each of the field effect transistors, and responsive to other of the field effect transistors experiencing an increase in current flow during the toggle and a difference in magnitudes of the currents exceeding a threshold, deactivate the field effect transistors.

6. The vehicle of claim **5**, wherein the one or more controllers are further programmed to, while the field effect transistors are passing currents, and responsive to other of the field effect transistors experiencing an increase in current flow during the toggle and the difference being less than the threshold, maintain operation of the field effect transistors.

7. The vehicle of claim **1**, wherein the one or more controllers are further programmed to generate an alert responsive to, for one of the field effect transistors, the electrical parameters resulting from the corresponding pulse being reported for more than one of the inputs.

8. The vehicle of claim **1**, wherein the electrical parameters are voltages.

9. A method comprising:

while a plurality of field effect transistors of a power distribution module of a vehicle are passing currents to an upfitter load, sequentially toggling each of the field effect transistors, and responsive to other of the field effect transistors experiencing an increase in current flow during the toggling, deactivating the field effect transistors.

10. The method of claim **9** further comprising, while the field effect transistors are connected with at least one input of an upfitter power module, sequentially pulsing the field effect transistors, and responsive to, for one of the field effect transistors, electrical parameters resulting from the corresponding pulsing being reported for more than one of the inputs, precluding operation of the upfitter power module.

11. The method of claim **10** further comprising, responsive to, for each of the field effect transistors, an electrical parameter resulting from the pulsing being reported for only one of the inputs, enabling the upfitter power module.

12. The method of claim **9** further comprising, while the field effect transistors are connected with at least one input of an upfitter power module, sequentially pulsing the field effect transistors, and responsive to, for two or more of the field effect transistors, electrical parameters resulting from the corresponding pulsing being reported for a same one of the inputs, precluding operation of the upfitter power module.

13. The method of claim **12** further comprising, responsive to, for each of the field effect transistors, an electrical parameter resulting from the pulsing being reported for only one of the inputs, enabling the upfitter power module.

14. The method of claim **9** further comprising, responsive to, for the one of the field effect transistors, electrical parameters resulting from the corresponding pulsing being reported for more than one of the inputs, generating an alert.

15. A power control system of a vehicle comprising:

one or more controllers programmed to, after activation of the vehicle, sequentially pulse field effect transistors of a power distribution module of the vehicle that are connected to inputs of an upfitter power module, and preclude operation of the upfitter power module responsive to, for at least two of the field effect transistors, electrical parameters resulting from the corresponding pulses being reported for a same one of the inputs.

16. The power control system of claim **15**, wherein the one or more controllers are further programmed to preclude operation of the upfitter power module responsive to, for one of the field effect transistors, electrical parameters resulting from the corresponding pulse being reported for more than one of the inputs.

17. The power control system of claim **15**, wherein the one or more controllers are further programmed to enable the upfitter power module responsive to, for each of the field effect transistors, an electrical parameter resulting from the pulse being reported for only one of the inputs.

18. The power control system of claim **15**, wherein the one or more controllers are further programmed to, while the field effect transistors are passing currents, sequentially toggle each of the field effect transistors, and responsive to other of the field effect transistors experiencing an increase in current flow during the toggle, deactivate the field effect transistors.

19. The power control system of claim **15**, wherein the one or more controllers are further programmed to generate an alert responsive to, for at least two of the field effect transistors, electrical parameters resulting from the corresponding pulses being reported for the same one of the inputs.

20. The power control system of claim **15**, wherein the electrical parameters are voltages.

* * * * *